

KL is the wrong divergence for importance sampling: a sharp factor-of-two for Gaussians

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Abstract

A popular rule of thumb, made rigorous by Chatterjee and Diaconis, says importance sampling needs about $e^{D(p\|q)}$ samples, where D is the Kullback–Leibler divergence of the target p from the proposal q . But KL controls the *typical* weight, not its variance, and the standard error bars and effective-sample-size (ESS) diagnostics live on the variance. The two functionals are the Rényi divergences of orders 1 and 2, so $D_1 \leq D_2$ always — but the gap can be infinite. We make this concrete with the cleanest possible example: for a Gaussian proposal $q = N(0, 1)$ and Gaussian target $p = N(0, \sigma^2)$, the importance weights have **finite variance if and only if** $\sigma^2 < 2$, with the exact closed form $1 + \chi^2(p\|q) = 1/(\sigma\sqrt{2 - \sigma^2})$, while $D(p\|q)$ stays below 0.16 nats all the way to the blow-up. A KL-based sample-size rule, and the Kish ESS, both report “healthy” exactly where the estimator’s error stops shrinking at the $1/\sqrt{n}$ rate. Simulations confirm the dichotomy and show that the threshold $\sigma^2 = 2$ is precisely the Pareto- $\hat{k} = \frac{1}{2}$ boundary used by modern smoothed importance sampling.

1. Setup and notation

Let q (the *proposal*) and p (the *target*) be densities on $\mathbb{R}^d$ with $p \ll q$, and let $w(x) = p(x)/q(x)$ be the importance weight. To estimate $\mathbb{E}_p[f] = \int f p$ from $x_1, \dots, x_n \stackrel{\text{iid}}{\sim} q$, self-normalized importance sampling (SNIS) uses

$$\widehat{\mathbb{E}_p[f]} = \frac{\sum_{i=1}^n w(x_i) f(x_i)}{\sum_{i=1}^n w(x_i)}. \quad (1)$$

Two divergences of p from q govern how well (1) behaves. Both are *forward* (target-over-proposal) Rényi divergences $D_\alpha(p\|q) = \frac{1}{\alpha-1} \log \mathbb{E}_q[w^\alpha]$:

$$D_1 = D(p\|q) = \mathbb{E}_q[w \log w], \quad D_2 = \log \mathbb{E}_q[w^2] = \log(1 + \chi^2(p\|q)). \quad (2)$$

D_1 is the ordinary KL divergence; D_2 is the log of the second moment of the weights, equivalently the log of $1 + \chi^2$. **Chatterjee and Diaconis (2018)** show that, in a precise sense, $n \asymp e^{D_1}$ samples are necessary and sufficient for SNIS to be accurate. But the textbook variance of (1), the width of any weight-based confidence interval, and the **Kish effective sample size**

$$\text{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}, \quad \frac{\text{ESS}}{n} \xrightarrow{\text{a.s.}} \frac{1}{\mathbb{E}_q[w^2]} = e^{-D_2}, \quad (3)$$

are all governed by D_2 , not D_1 . The question of this note: how far apart can D_1 and D_2 be? The answer is *arbitrarily*, and a one-dimensional Gaussian already shows it.

2. Results

2.1 KL never exceeds the second-moment exponent

Proposition 1. *For any $p \ll q$, $D_1(p \parallel q) \leq D_2(p \parallel q)$, i.e. $D(p \parallel q) \leq \log(1 + \chi^2(p \parallel q))$.*

Proof. This is monotonicity of Rényi divergence in α (van Erven and Harremoës, 2014). A self-contained one line: $w \geq 0$ with $\mathbb{E}_q[w] = 1$, so $d\mu = w dq$ is a probability measure, and by Jensen’s inequality (concavity of $\log$),

$$D_1 = \mathbb{E}_q[w \log w] = \mathbb{E}_\mu[\log w] \leq \log \mathbb{E}_\mu[w] = \log \mathbb{E}_q[w^2] = D_2. \quad \square$$

So the Chatterjee–Diaconis budget e^{D_1} is always *no larger* than the variance budget e^{D_2} : KL is an optimistic surrogate. Equality of orders ($D_2 = \Theta(D_1)$) holds when the log-weights are light-tailed; the next two results show it can fail completely.

2.2 A Gaussian scale family: the threshold is exactly $\sigma^2 = 2$

Fix $q = N(0, 1)$ and $p = N(0, \sigma^2)$.

Proposition 2. *The second moment of the importance weights is*

$$\mathbb{E}_q[w^2] = 1 + \chi^2(p \parallel q) = \begin{cases} \frac{1}{\sigma\sqrt{2-\sigma^2}}, & \sigma^2 < 2, \\ +\infty, & \sigma^2 \geq 2. \end{cases} \quad (4)$$

Consequently the weights have finite variance — and SNIS enjoys an asymptotic $1/\sqrt{n}$ rate with finite asymptotic variance — if and only if $\sigma^2 < 2$. Meanwhile

$$D(p \parallel q) = \frac{1}{2}(\sigma^2 - 1 - \log \sigma^2) \leq \frac{1}{2}(1 - \log 2) \approx 0.1534 \text{ nats for } \sigma^2 \leq 2. \quad (5)$$

Proof. With $p(x)^2/q(x) = \frac{1}{\sigma^2\sqrt{2\pi}} \exp(-x^2(\frac{1}{\sigma^2} - \frac{1}{2}))$, the integral $\int p^2/q$ converges iff the coefficient $a = \frac{1}{\sigma^2} - \frac{1}{2} = \frac{2-\sigma^2}{2\sigma^2}$ is positive, i.e. iff $\sigma^2 < 2$. When it is, $\int e^{-ax^2} dx = \sqrt{\pi/a}$ gives $\int p^2/q = \frac{1}{\sigma^2\sqrt{2\pi}} \sqrt{\pi/a} = 1/(\sigma\sqrt{2-\sigma^2})$. The KL formula is the standard one for two centered Gaussians, and $\sigma^2 - 1 - \log \sigma^2$ increases on $[1, 2]$, with value $1 - \log 2$ at $\sigma^2 = 2$. $\square$

The picture is stark (Figure 1): as $\sigma^2 \uparrow 2$, χ^2 (hence D_2) diverges to $+\infty$, while D_1 glides smoothly through 0.153 as if nothing were happening. A practitioner monitoring KL — or running a variational procedure that *minimizes* KL — sees a target a fraction of a nat away from the proposal and concludes a handful of samples suffice. The estimator’s variance is, in fact, infinite.

2.3 Location vs. scale: where the gap comes from

Why does KL miss this? The gap $D_2 - D_1$ is a statement about the *tails* of the weight, and a location shift leaves Gaussian tails intact while a scale change does not.

Proposition 3. *For the location family $q = N(0, 1)$, $p = N(\mu, 1)$,*

$$D_2 = \mu^2 = 2D_1 \quad (\text{both finite for all } \mu), \quad (6)$$

Same proposal, same target family: KL is blind to the variance blow-up

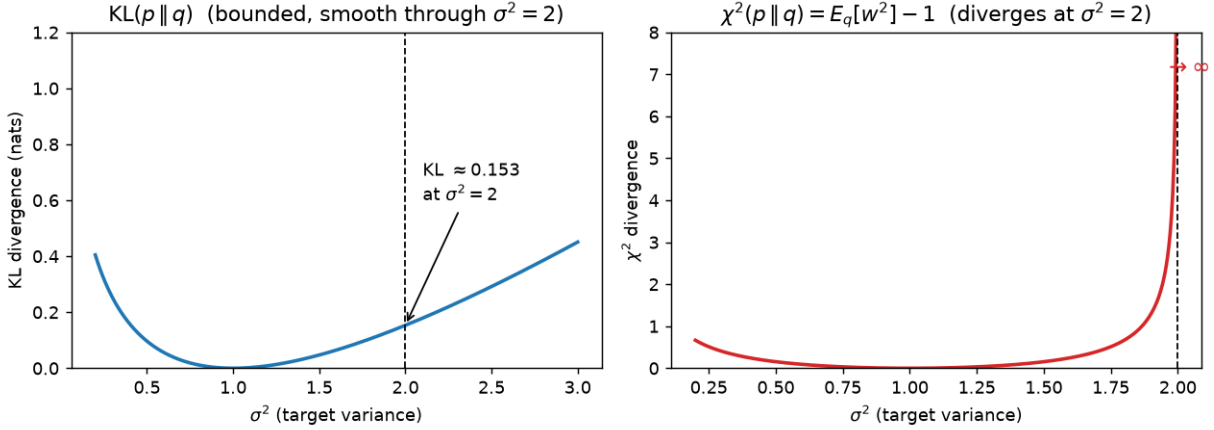


Figure 1: Figure 1. Same proposal $q = N(0, 1)$, same target family $N(0, \sigma^2)$. Left: KL is bounded and smooth across the threshold. Right: $\chi^2 = e^{D_2} - 1$ diverges at $\sigma^2 = 2$.

so KL and the second-moment exponent agree up to a factor of two. For the scale family of Proposition 2, $D_1 \leq 0.153$ while $D_2 \rightarrow \infty$ as $\sigma^2 \uparrow 2$ — an unbounded ratio.

Proof. For the location family, $\int p^2/q = e^{\mu^2}$ by completing the square, so $D_2 = \mu^2$; and $D_1 = \mu^2/2$ is the Gaussian mean-shift KL. The scale claims are Proposition 2. $\square$

So in the regime where KL was *designed* to be informative — matching a location, the implicit job of a mean-field or Laplace proposal — it tracks the variance budget. The moment the mismatch is one of *spread* (proposal lighter-tailed than target), KL and D_2 part ways, and only D_2 tells the truth.

Remark (the factor-of-two is the Pareto- $\frac{1}{2}$ boundary). The weight $w(x) \propto \exp(\frac{1}{2}(1 - \frac{1}{\sigma^2})x^2)$ has, under $x \sim q$, a power-law upper tail with index $\sigma^2/(\sigma^2 - 1)$, i.e. a generalized-Pareto shape $\hat{k} = 1 - 1/\sigma^2$. Finite variance ($\hat{k} < \frac{1}{2}$) is exactly $\sigma^2 < 2$; finite mean ($\hat{k} < 1$) holds for all σ^2 . Thus the threshold $\sigma^2 = 2$ is the $\hat{k} = \frac{1}{2}$ line at which Pareto-smoothed importance sampling (Vehtari et al., 2024) flags the weights as unreliable — a reassuring cross-check from a completely different diagnostic.

3. Simulations

`sim.py` (seed 20260625, NumPy) fixes $q = N(0, 1)$ and estimates $\mathbb{E}_p[X^2] = \sigma^2$ by SNIS (1). All numbers below are its actual output.

Closed form (4) holds. Monte-Carlo $\mathbb{E}_q[w^2]$ over 2×10^7 draws: at $\sigma^2 = 0.5, 1.0, 1.5$ we get 1.1545, 1.0000, 1.1523 against the formula’s 1.1547, 1.0000, 1.1547. At $\sigma^2 = 1.8$ the estimate (1.549) already *undershoots* the truth (1.667): even estimating the second moment becomes hard as the boundary nears, foreshadowing the blow-up.

The error stops shrinking at $\sigma^2 = 2$ (Figure 2). RMSE of the SNIS estimate over 400 replications, from $n = 100$ to $n = 10^5$ (a $1000\times$ increase, for which a clean $1/\sqrt{n}$ rate would shrink RMSE by $\sqrt{1000} \approx 31.6$):

σ^2	weights	RMSE @ $n=100$	RMSE @ $n=10^5$	shrink factor
1.5	finite var	0.379	0.015	25.1
2.5	infinite var	1.300	0.137	9.5
3.0	infinite var	1.474	0.583	2.5

The finite-variance case nearly attains the ideal 31.6; the infinite-variance cases fall badly short, with $\sigma^2 = 3$ barely improving over three orders of magnitude in n . (SNIS remains *consistent* throughout — numerator and denominator have finite means — so this is a failure of *rate*, driven by rare enormous weights, not of bias.)

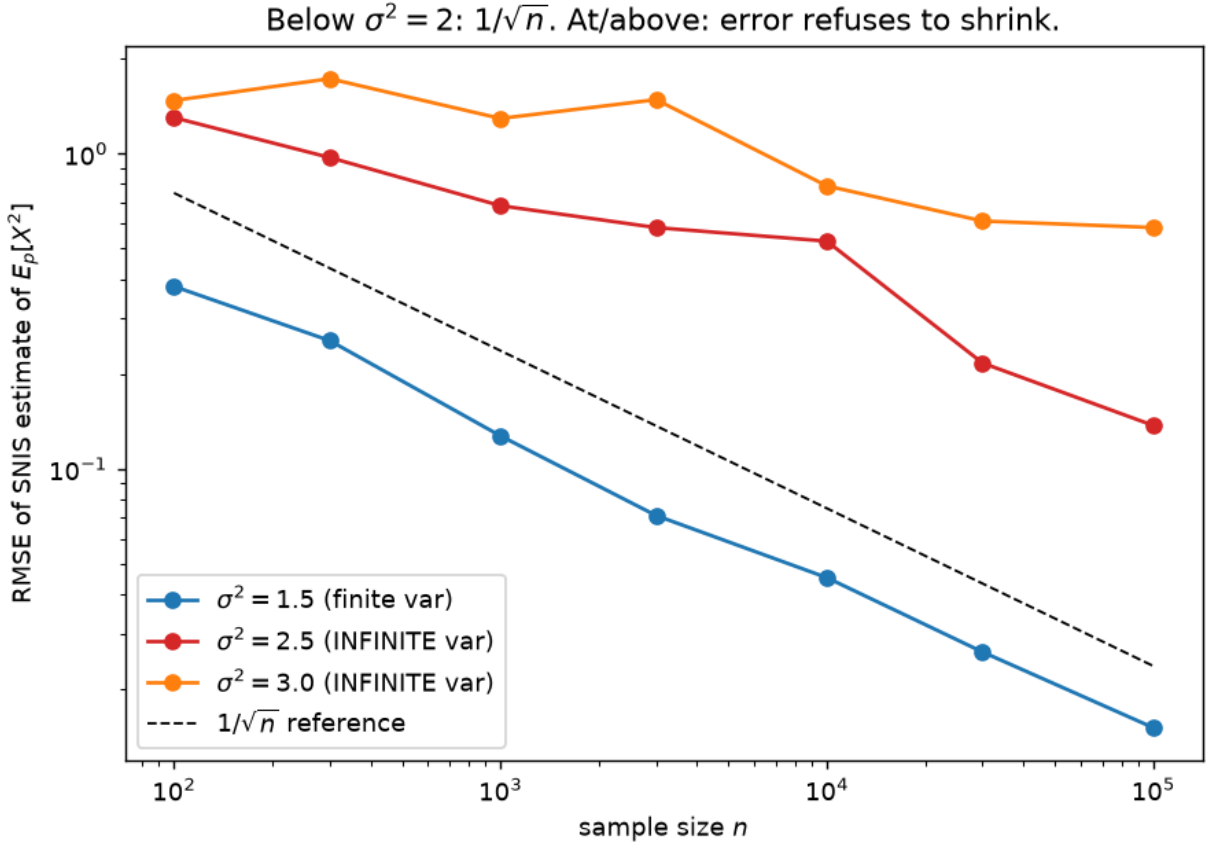


Figure 2: Figure 2. RMSE of the SNIS estimate of $E_p[X^2]$ vs. sample size, log-log. Below $\sigma^2 = 2$ the error tracks $1/\sqrt{n}$; at and above it, the error refuses to shrink.

The ESS diagnostic lies (Figure 3). Over 3000 runs at $n = 5000$, the Kish relative ESS (3) for the *infinite-variance* target $\sigma^2 = 3$ has mean 0.231 and median 0.234 — a “moderate” reading a practitioner would tolerate — even though the true relative ESS e^{-D_2} is 0. The finite-variance $\sigma^2 = 1.5$ sits reassuringly at mean 0.869. The ESS estimator simply cannot see a second moment that does not exist; it reports the second moment of the *sample*, which is always finite. A long single run ([running_estimate.png](#)) makes the mechanism visible: smooth convergence at $\sigma^2 = 1.5$, versus jump-driven wandering at $\sigma^2 = 2.5, 3.0$ as occasional huge weights reset the estimate.

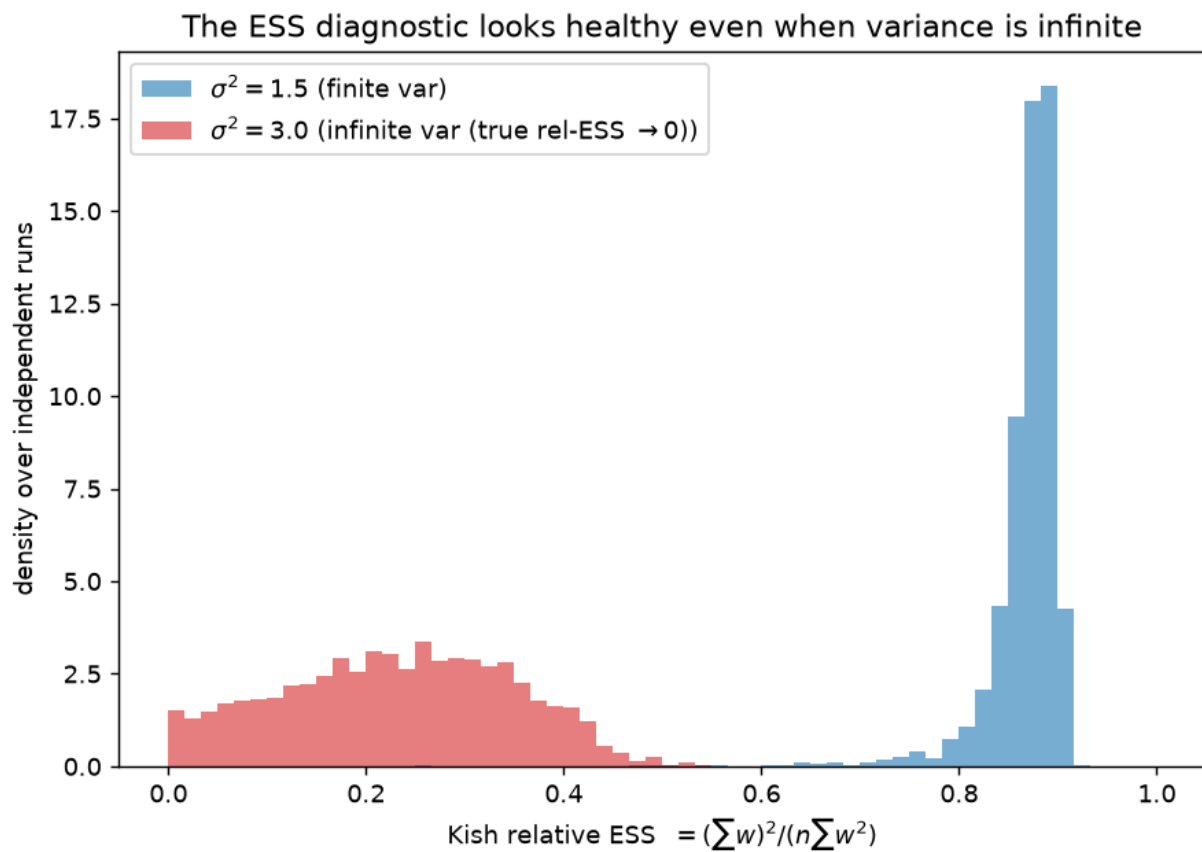


Figure 3: Figure 3. Distribution of the Kish relative ESS over independent runs ($n = 5000$). For $\sigma^2 = 3$ (infinite variance, true rel-ESS = 0) the diagnostic still clusters near 0.23 — false confidence.

4. Discussion

The moral is a one-liner: **importance sampling is controlled by χ^2 (Rényi-2), not by KL (Rényi-1), and Proposition 1 makes KL a systematically optimistic surrogate.** This matters because KL is the divergence we reach for by default — it is what variational inference minimizes and what the Chatterjee–Diaconis sample-size heuristic is phrased in. A proposal can be excellent by KL and useless by variance; the Gaussian scale family pins the failure to a single, memorable number, $\sigma^2 = 2$, and the cleanest possible closed form (4). The same number is the Pareto- $\hat{k} = \frac{1}{2}$ threshold, which is encouraging: the practical fix already in use — Pareto-smoothed importance sampling and its $\hat{k}$ diagnostic (Vehtari et al., 2024), or simply monitoring an estimate of χ^2 / the second moment rather than KL or raw ESS — is detecting exactly the quantity the theory says matters.

None of the ingredients is individually new: the finite-variance condition for Gaussian importance sampling is folklore (e.g. Owen, 2013); the centrality of χ^2 and the second moment is the theme of Agapiou et al. (2017); the KL sample-size law is Chatterjee and Diaconis (2018); the unreliability of ESS is argued by Elvira et al. (2022). The contribution here is the *synthesis as a sharp separation*: the exact threshold (4), the Rényi-order reading $D_1 \leq D_2$ that explains why KL must be optimistic, and the location-vs-scale dichotomy (Proposition 3) that localizes the failure to tail mismatch — illustrated by a simulation in which a KL-based rule and the ESS diagnostic both certify a divergent estimator as healthy.

Limitations. (i) The dramatic version of the gap ($D_2 = \infty$, D_1 small) is special to families where the proposal’s tails are lighter than the target’s; for many realistic proposals D_2 is finite but merely large, and then KL and χ^2 differ by a factor, not qualitatively — though Proposition 1 still says KL understates the cost. (ii) Chatterjee and Diaconis study a genuinely different and legitimate notion of accuracy (typical relative error of the *unnormalized* estimator / normalizing constant), for which e^{D_1} is the right answer; our point is not that their theorem is wrong but that its functional D_1 does not control the variance/ESS that practitioners read off, and the two can diverge. (iii) Everything is one-dimensional; in d dimensions the second moment of a product proposal/target multiplies across coordinates, so χ^2 compounds geometrically and the gap to KL can only widen, but we have not quantified the multivariate case here. (iv) SNIS stays consistent in all our regimes; the failure is of rate and of error-bar honesty, not of point-estimate validity.

References

1. S. Chatterjee and P. Diaconis. “The sample size required in importance sampling.” *Annals of Applied Probability* 28(2):1099–1135, 2018. arXiv:1511.01437.
2. S. Agapiou, O. Papaspiliopoulos, D. Sanz-Alonso, and A. M. Stuart. “Importance Sampling: Intrinsic Dimension and Computational Cost.” *Statistical Science* 32(3):405–431, 2017. arXiv:1511.06196.
3. A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and J. Gabry. “Pareto Smoothed Importance Sampling.” *Journal of Machine Learning Research* 25(72):1–58, 2024. arXiv:1507.02646.
4. V. Elvira, L. Martino, and C. P. Robert. “Rethinking the Effective Sample Size.” *International Statistical Review* 90(3):525–550, 2022. arXiv:1809.04129.
5. T. van Erven and P. Harremoës. “Rényi Divergence and Kullback–Leibler Divergence.” *IEEE Transactions on Information Theory* 60(7):3797–3820, 2014. arXiv:1206.2459.
6. A. B. Owen. *Monte Carlo Theory, Methods and Examples*. 2013. (Ch. 9, importance sampling; finite-variance conditions.)